

aspects of life, there may arise a need to send an SMS via the Arduino™ shield to your mobile phone to monitor many conditions and take actions depending on that condition such as the thirsty plant in need of water that we saw in Chapter 1. Instead of email, SMS could have been a medium for communication. Critical application areas such as healthcare are better suited to use SMS in case of emergencies even when Internet might be available, since there's a higher possibility of an SMS being sent.

Let's look at the code to send an SMS using an Arduino™ device.

You can find a copy of the code <http://dgit.in/ardsendsmscode> at and related explanation at <http://dgit.in/ardsendsmsexp>.

Connect to Internet via GPRS

Say, a Wi-Fi network is unavailable for use, such as when travelling, what do you do in such a case? During such times, you can program the Arduino™ board to switch over to a cellular GPRS network for internet connectivity.

When using data services, the electrical current requirement of the GSM shield may increase to 2 A. This makes it absolutely necessary to connect the shield to an external power source, else you might end up damaging the Arduino™ board and the shield and probably some other connected hardware.

You can find a copy of the code at <http://dgit.in/ardgprscode> and related explanation at <http://dgit.in/ardgprsexp>.

NOTE:

When using a shield, proper power supply should be connected to supply the required power. Though the Arduino™ board can be powered via the USB port of your computer, the shields you're using or the components connected to them require much more power for smooth operation. For instance: The GSM shield may need up to 2 A of current for proper operation, but the USB port of a computer can provide a maximum of 500mA of current via the USB 2.0 port and 900mA current via the newer USB 3.0 port.

Syntax for writing Arduino™ sketches

To write any Arduino™ sketch, you need to be aware of the syntax used for writing the sketch. When you know the syntax, writing a program is just like fitting the pieces of puzzle correctly together. In this section, we'll look at the programming syntax in Arduino™ that's an add-on to regular C language.